

Fig. 7 — The matched spectral-versus-colour comparison: one dataset, one model, one flag, five seeds

HistologyHSI-BC, 35/5/5 patient-disjoint split, --rgb_source synthetic. The two runs differ in exactly one command-line argument (--data_dir).

What is held fixed

Same cubes

35 train / 5 val / 5 test patients, no leakage identical 348,894-patch test split

Same weights budget

446,409 trainable parameters in BOTH arms — the structural band-agnosticism result

Same recipe

bs 256, bf16, 20 epochs, AdamW 3e-4, wd 0.05, focal-weighted $\gamma 1.5$ ($p=0.75$), step-axis cosine, EMA .9995

Same recursion

d = 128, 2 core layers, 6 latent · 3 improve · 3 segments = 63 core applications, mlp mixer

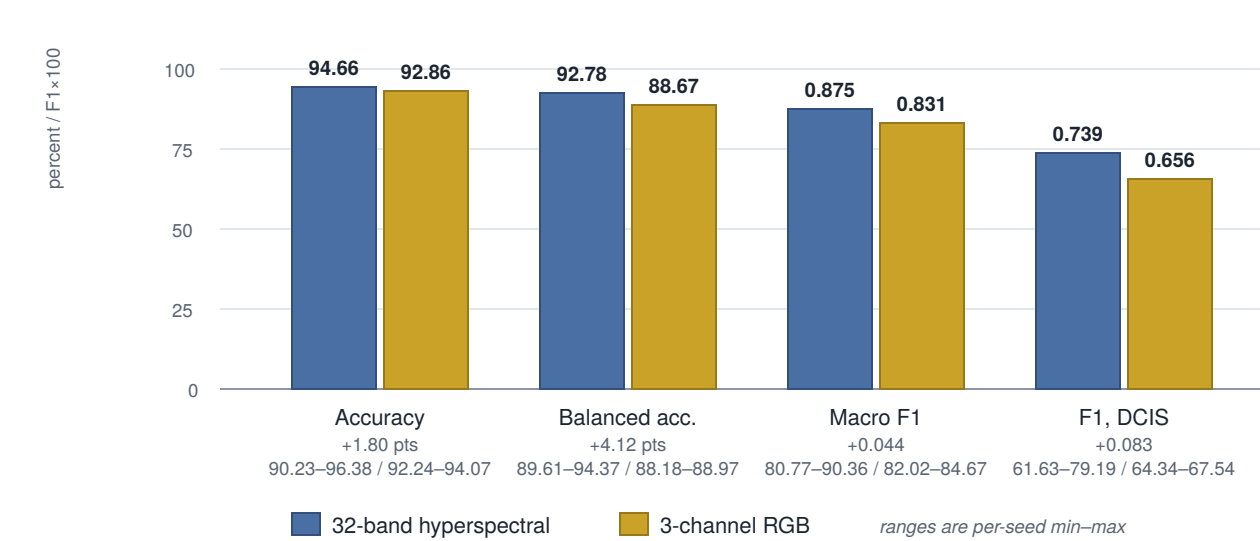
What differs

C = 32 bands vs C = 3 channels

Five seeds

1, 7, 13, 23, 42 — both arms, all 20 epochs.
 Δ balanced acc. positive at 5/5; Δ macro-F1 at 4/5

Held-out test, 348,894 patches from five patients used for neither training nor selection — mean over five seeds



Where the difference lives

Per-class F1, mean over five seeds



Both arms solve IDC — 0.993 vs 0.994, the one class where RGB is marginally ahead. The margin is the healthy/DCIS boundary, and most of it is DCIS: +0.083 mean, the 7 % minority class and the clinically consequential call.

Ranking and calibration, seed 42

macro ROC-AUC 0.9957 vs 0.9907
macro PR-AUC 0.9488 vs 0.9180
ECE 0.0565 vs 0.0903
Cohen's κ 0.923 vs 0.839
so the gap is not an artefact of one decision threshold.

Read this way

Matched on every axis the pipeline records, so the spectral axis is the only intentional difference. Balanced accuracy favours hyperspectral at all five seeds (+4.12 ± 1.66 pts, sem 0.74); macro-F1 at four of five (+0.044 ± 0.038), inverting at seed 1, where RGB leads by 0.013. The hyperspectral arm is also the less stable of the two: its balanced accuracy spans 4.8 points across seeds against the RGB arm's 0.8. Five test patients, one dataset, no significance test. The claim licensed is "on this dataset, under this protocol", not a claim about HSI in general.